Questions

Q1.

The line l_1 has equation

$$10x - 2y + 7 = 0$$

(a) Find the gradient of l_1

(1)

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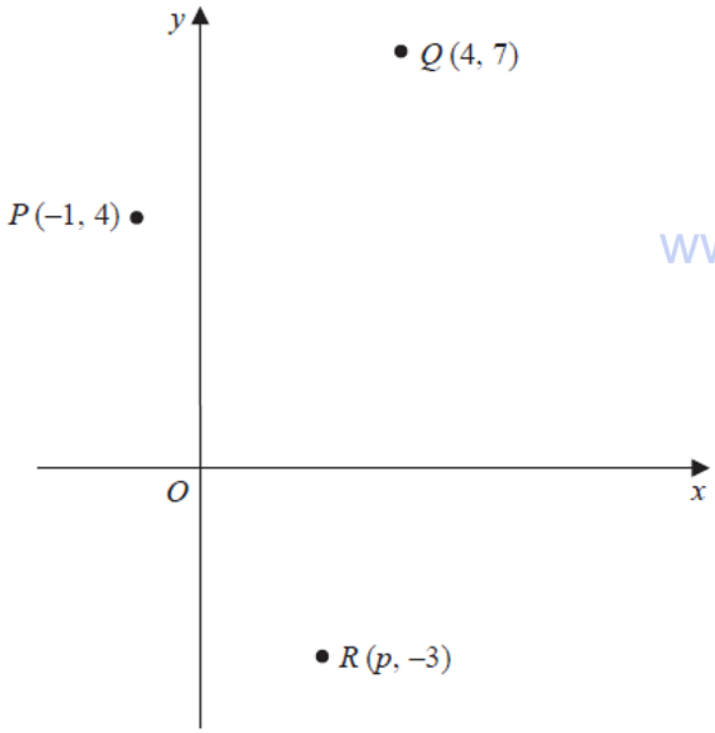
The line l_2 is parallel to the line l_1 and passes through the point $\left(-\frac{1}{3}, \frac{4}{3}\right)$.

(b) Find the equation of l_2 in the form $y = mx + c$, where m and c are constants.

(3)

(Total for question = 4 marks)

Q2.



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Figure 3

Figure 3 shows the points P, Q and R.

Points P and Q have coordinates $(-1, 4)$ and $(4, 7)$ respectively.

(a) Find an equation for the straight line passing through points P and Q.

Give your answer in the form $ax + by + c = 0$, where a, b and c are integers.

(4)

The point R has coordinates $(p, -3)$, where p is a positive constant.

Given that angle $QPR = 90°$,

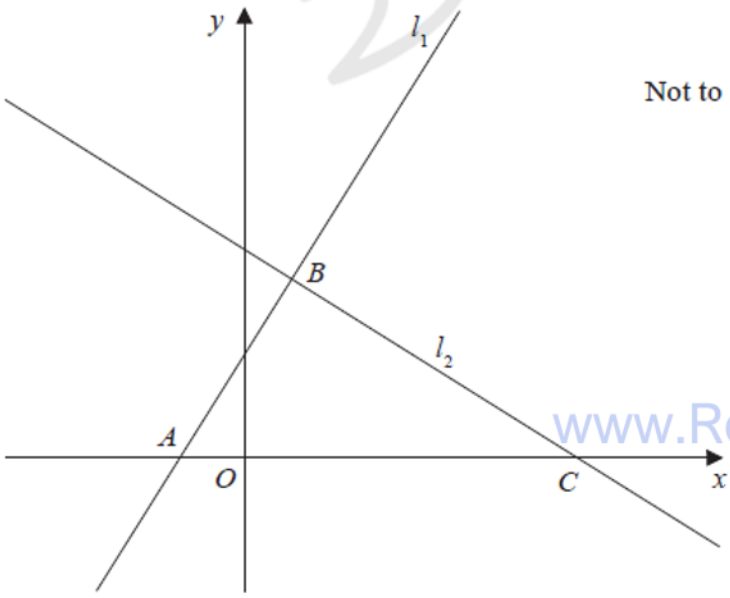
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(b) find the value of p.

(3)

(Total for question = 7 marks)

Q3.



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Figure 2

The straight line l_1 has equation $2y = 3x + 5$

The line l_1 cuts the x-axis at the point A, as shown in Figure 2.

(a) (i) State the gradient of l_1

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(ii) Write down the x coordinate of point A.

(3)

Another straight line l_2 intersects l_1 at the point B with x coordinate 1 and crosses the x-axis at the point C, as shown in Figure 2.

Given that l_2 is perpendicular to l_1

(b) find an equation for l_2 in the form $ax + by + c = 0$, where a, b and c are integers,

(5)

(c) find the exact area of triangle ABC.

(3)

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(Total for question = 11 marks)

Q4.

The line l_1 has equation $2x + 3y = 6$

The line l_2 is parallel to the line l_1 and passes through the point $(3, -5)$.

Find the equation for the line l_2 in the form $y = mx + c$, where m and c are constants.

(4)

(Total for question = 4 marks)

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Q5.

The line l_1 has equation $x + 3y - 11 = 0$

The point A and the point B lie on l_1

Given that A has coordinates $(-1, p)$ and B has coordinates $(q, 2)$, where p and q are integers,

(a) find the value of p and the value of q,

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(2)

(b) find the length of AB, giving your answer as a simplified surd.

(2)

The line l_2 is perpendicular to l_1 and passes through the midpoint of AB.

(c) Find an equation for l_2 giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(5)

(Total for question = 9 marks)

Q6.

A line l passes through the points A (5, –2) and B (1, 10).

Find the equation of l, writing your answer in the form $y = mx + c$ where m and c are constants.

(Total for question = 3 marks)

Q7.

The line l_1 passes through the points $A(-1, 4)$ and $B(5, -8)$

(a) Find the gradient of l_1

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(2)

The line l_2 is perpendicular to the line l_1 and passes through the point $B(5, -8)$

(b) Find an equation for l_2 in the form $ax + by + c = 0$, where a, b and c are integers.

(4)

(Total for question = 6 marks)

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Q8.

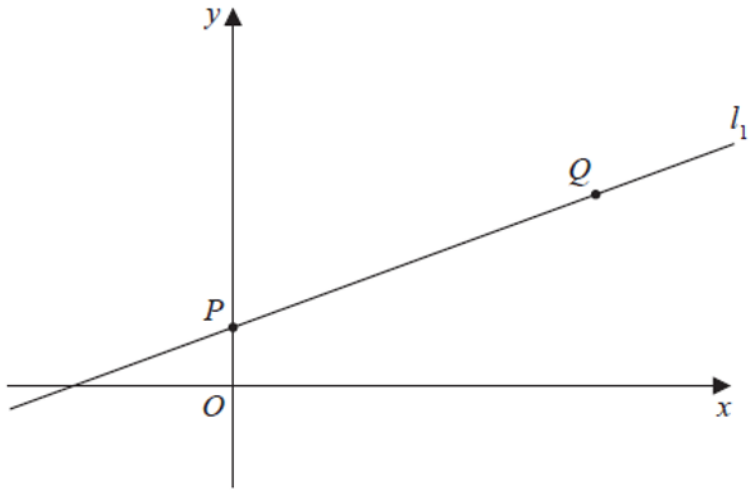


Figure 1

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The line l_1, shown in Figure 1, has equation $2y = 3x + 8$

The line l_1 intersects the y-axis at the point P and passes through the point Q with x coordinate 6

(a) Find (i) the coordinates of P,

(ii) the coordinates of Q.

(2)

The line l_2 is perpendicular to l_1 and passes through the point Q.

(b) Find an equation for l_2, writing the answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(4)

The line l_2 cuts the x-axis at the point R.

(c) Find the area of quadrilateral $OPQR$, making your method clear.

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(3)

(Total for question = 9 marks)

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